INTRODUCTION

Project Overview

The project titled **"Comprehensive Analysis and Dietary Strategies – A College Food Choices Case Study"** focuses on understanding and analyzing the dietary habits of college students. In today's fast-paced academic environment, students often struggle to maintain balanced nutrition due to stress, limited budgets, irregular schedules, and emotional eating triggers.

This project aims to bridge the gap between students' *ideal dietary goals* and their *actual food consumption patterns*. By collecting and analyzing data related to calorie intake, BMI, emotional triggers, food categories, and lifestyle behaviors, the system provides meaningful insights into students' eating habits.

The solution integrates data prepossessing, interactive dashboard visualization, and AI-powered assistance to support better decision-making. The developed dashboard allows users to monitor calorie intake, nutritional balance, and health indicators through clear visual representations such as bar charts, line graphs, and heat maps. Additionally, the AI-based wellness assistant provides personalized guidance to promote healthier lifestyle choices.

Overall, this project combines data analytics and intelligent systems to improve nutritional awareness and encourage balanced dietary practices among college students.

Purpose

The primary purpose of this project is to analyze existing dietary behaviors of college students and develop effective strategies to improve their overall health and well-being.

The specific objectives of the project are:

- To study common food consumption patterns among college students.
- To identify nutritional deficiencies and unhealthy eating trends.
- To analyze the impact of emotional and environmental triggers on food choices.
- To calculate key health indicators such as BMI and average calorie intake.
- To design an interactive dashboard for visualizing dietary data.
- To develop an AI-powered wellness assistant that provides personalized dietary recommendations.

Through this system, students can gain better awareness of their eating habits and make data-driven decisions toward achieving a balanced and healthy diet. The project also supports educational institutions in promoting student wellness initiatives and long-term health improvement.

Ideation Phase

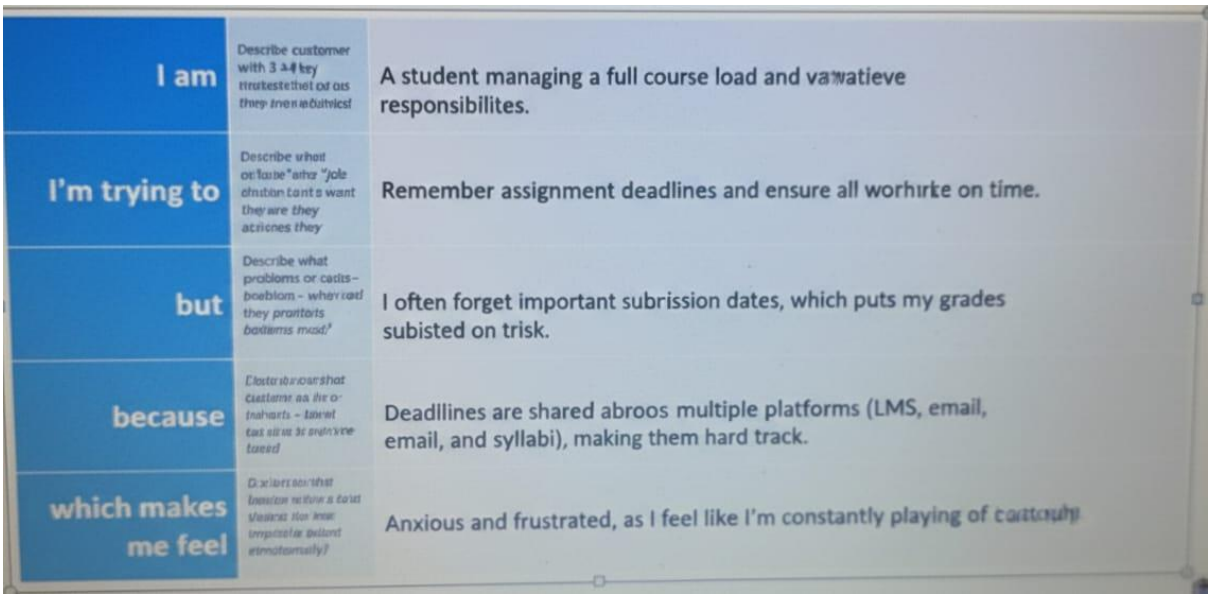
Define the Problem Statements

Date	31 January 2025
Team ID	LTVIP2026TMIDS72995
Project Name	Comprehensive Analysis and Dietary Strategies A College Food Choices Case Study
Maximum Marks	2 Marks

Customer Problem Statement Template:

Create a problem statement to understand your customer's point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love.

A well-articulated customer problem statement allows you and your team to find the ideal solution for the challenges your customers face. Throughout the process, you'll also be able to empathize with your customers, which helps you better understand how they perceive your product or service.



I am	Describe customer with 3-4 key	A student managing a full course load and va'watieve responsibilites.
I'm trying to	Describe what	Remember assignment deadlines and ensure all worhirke on time.
but	Describe what problems or cells–boeblom – wher'ced they prontorts	I often forget important subrission dates, which puts my grades subisted on trisk.
because		Deadllines are shared abroos multiple platforms (LMS, email, email, and syllabi), making them hard track.
which makes me feel		Anxious and frustrated, as I feel like I'm constantly playing of cartcuhp

Ideation Phase

Empathize & Discover

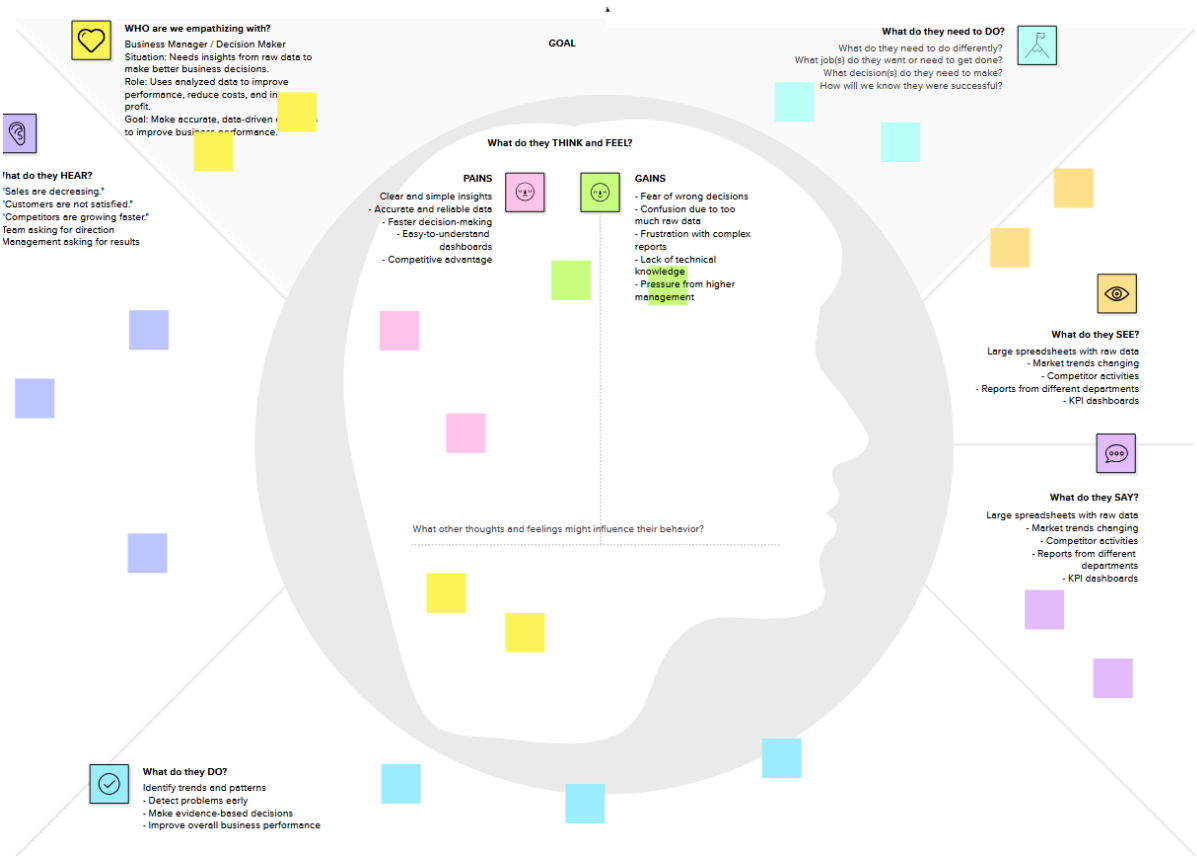
Date	31 January 2025
Team ID	LTVIP2026TMIDS72995
Project Name	Comprehensive Analysis and Dietary Strategies A College Food Choices Case Study
Maximum Marks	4 Marks

Empathy Map Canvas:

An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviours and attitudes.

It is a useful tool to helps teams better understand their users.

Creating an effective solution requires understanding the true problem and the person who is experiencing it. The exercise of creating the map helps participants consider things from the user's perspective along with his or her goals and challenge



Ideation Phase

Brainstorm & Idea Prioritization Template

Date	3 february 2026
Team ID	LTVIP2026TMIDS72995
Project Name	**Comprehensive Analysis and Dietary Strategies A College Food Choices Case Study**
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization Template:

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions.

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

Step-1: Team Gathering, Collaboration and Select the Problem Statement

The team gathered and discussed various health and nutrition-related issues observed in daily life. Through collaborative discussion, the team identified improper dietary habits and lack of nutritional awareness as major contributors to lifestyle-related health problems.

Problem Statement:
To analyze existing dietary habits and develop effective dietary strategies to improve overall health and prevent lifestyle-related diseases.

Step-2: Brainstorm, Idea Listing and Grouping

Ideas generated during brainstorming include:
- Study of common dietary patterns
- Identification of nutritional deficiencies
- Creation of balanced and affordable diet plans
- Awareness on calorie intake and portion control
- Dietary strategies for weight management and general health

Grouping of Ideas:
1. Nutritional Analysis
2. Dietary Planning
3. Health Improvement Strategies
4. Awareness and Education

Step-3: Idea Prioritization

Based on feasibility and impact, the team prioritized the development of practical dietary strategies supported by nutritional analysis. The final idea focuses on promoting balanced nutrition using simple, accessible, and cost-effective food choices.

Customer Journey Map: College Student Food Choices

	Trigger (Stress/Hunger)	Planning (Budget/Time)	Decision (Purchase)	Consumption	Reflection (Goal Setting)
Steps	Feels stress, boredom, or hunger.	Assessing convenience vs. health.	Choosing dining hall or fast food.	Eating "college diet" (pizza, snacks).	Comparing to "ideal diet" goals.
Goals & Motivations	Find comfort or quick energy.	Find "cheap and easy" foods.	Convenience and low cost.	Quick satiety.	Weight loss, more veggies.
Touchpoints	Dorm, Library, Phone	Dining menus, Apps	Laker/Egan Halls, Fast Food	Desk, Dining Area	Health apps, Scale
Pain Points	Stress/Boredom/Anger triggers.	Limited time to cook.	Expensive eating out.	Not eating "ideal" fresh foods.	Negative self-perception of weight.
Opportunities	Healthy snack kits in study areas.	Pre-planned weekly meal guides.	Affordable healthy campus options.	Nutritional labels on all meals.	Student wellness/nutrition coaching.

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	10 february 2025
Team ID	LTVIP2026TMIDS72995
Project Name	Comprehensive Analysis and Dietary Strategies A College Food Choices Case Study
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

NFR No.	Requirement Type	Description
NFR-1	Usability	Measures the ease of use and user interface intuitiveness.
NFR-2	Security	Defines protection against unauthorized access and data integrity.
NFR-3	Reliability	The ability of the system to perform its functions under stated conditions.
NFR-4	Performance	Responsiveness and speed of the system under a given workload.
NFR-5	Availability	The percentage of time the system is operational and accessible.
NFR-6	Scalability	The system's ability to handle growing amounts of work or users.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Requirement Type	Description
NFR-1	Usability	Measures the ease of use and user interface intuitiveness.
NFR-2	Security	Defines protection against unauthorized access and data integrity.
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NFR-6	Scalability	The system's ability to handle growing amounts of work or users.

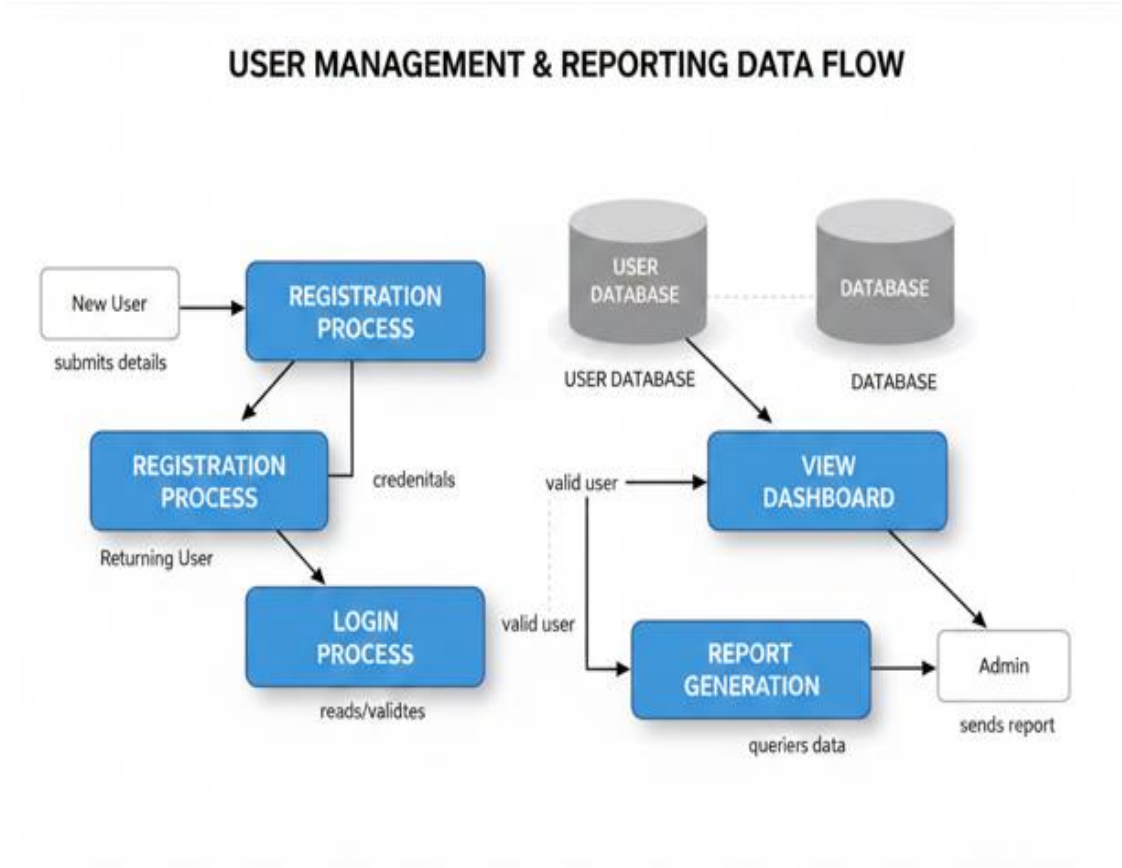
Project Design Phase-II

Data Flow Diagram & User Stories

Date	31 february 2025
Team ID	LTVIP2026TMIDS72995
Project Name	Comprehensive Analysis and Dietary Strategies A College Food Choices Case Study
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored



User Type	Epic	User Story ID	User Story	Acceptance Criteria	Priority	Release
Customer (Mobile)	Registration	USN-1	As a user, I can register using email and password.	User can successfully register and access dashboard.	High	Sprint-1
Customer (Mobile)	Email Confirmation	USN-2	As a user, I receive confirmation email after registration.	User receives and verifies email successfully.	High	Sprint-1
Customer (Mobile)	Login	USN-3	As a user, I can login using email and password.	User is authenticated and redirected to dashboard.	High	Sprint-1
Customer (Web)	Dashboard	USN-4	As a web user, I can view my dashboard data.	Dashboard displays user information correctly.	Medium	Sprint-2
Admin	User Management	USN-5	As an admin, I can manage user accounts.	Admin can view, edit, and delete user records.	High	Sprint-1
Customer Care Executive	Support	USN-6	As a support executive, I can respond to user queries.	Queries are updated and marked as resolved.	Medium	Sprint-2

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	13february 2026
Team ID	LTVIP2026TMIDS72995
Project Name	Comprehensive Analysis and Dietary Strategies A College Food Choices Case Study
Maximum Marks	4 Marks

Technical Architecture

Table-1: Components & Technologies

Following the standardized template, here is the mapping of the architecture components:

S.No	Component	Description	Technology
1.	User Interface	How the user interacts with the app (Web/Mobile).	React.js / Flutter
2.	Application Logic-1	Core logic for registration and login processes.	Python (FastAPI)
3.	Application Logic-2	Logic for handling natural language meal logs.	IBM Watson STT service
4.	Application Logic-3	Logic for interactive user guidance and support.	IBM Watson Assistant
5.	Database	Structured local storage for user profile data.	PostgreSQL / MySQL
6.	Cloud Database	Scalable database service for cloud deployment.	IBM Cloudant / DB2
7.	File Storage	Storage for user-uploaded photos and media.	IBM Cloud Object Storage
8.	External API-1	API for nutritional data validation.	Nutritionix API
9.	External API-2	API for secure external authentication.	Google / LinkedIn

S.No	Component	Description	Technology
			API +1
10.	ML Model	Model for predicting dietary improvements.	Scikit-learn Model
11.	Infrastructure	Deployment environment for the application.	Kubernetes / Docker

Table-2: Application Characteristics

The following non-functional traits ensure the robustness of the architecture:

S.No	Characteristics	Description	Technology
1.	Open-Source	Use of community-driven frameworks.	FastAPI, PostgreSQL
2.	Security	Implementation of access controls and encryptions.	SHA-256, OAuth 2.0
3.	Scalability	Support for 3-tier or microservices scaling.	Docker Containerization
4.	Availability	Ensuring the app remains online via load balancing.	Kubernetes Orchestration
5.	Performance	Considerations for requests per second and caching.	Redis Caching

PROJECT DESIGN PHASE-I

1. CUSTOMER SEGMENTS (CS) Identify exactly who the customer is. Example: Working parents with young children.	**2. CUSTOMER CONSTRAINTS (CC)** Limits such as budget, lack of cash, poor network, limited devices.	**3. BEHAVIOUR (BE)** Current online and offline actions customers take to solve the problem.
4. JOBS-TO-BE-DONE / PROBLEMS (J&P) Specific tasks customers try to complete and problems they face.	**5. AVAILABLE SOLUTIONS (AS)** Past solutions tried. Evaluate pros & cons of alternatives.	**6. PROBLEM ROOT CAUSE (RC)** Real reason the problem exists. Back story behind the need.
7. TRIGGERS (TR) Events that push customers to act now. Example: News, peer influence.	**8. EMOTIONS (EM)** How customers feel before and after. Example: Lost → Confident.	**9. THE SOLUTION (SL)** New Business: Keep blank until validated. Existing Business: Write and test fit.

Project Design Phase-I
Proposed Solution Template

Date	19february 2026
Team ID	LTVIP2026TMIDS72995
Project Name	Comprehensive Analysis and Dietary Strategies A College Food Choices Case Study
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	The primary challenge is the misalignment between college students' "ideal diet" and their actual consumption habits due to emotional and environmental triggers.
2.	Idea / Solution description	The proposed solution is an AI-powered Wellness Assistant that intercepts emotional eating triggers with personalized, healthy interventions.
3.	Novelty / Uniqueness	The solution is built on a robust 3-tier architecture to ensure reliability and performance.
4.	Social Impact / Customer Satisfaction	Improved Nutritional Alignment**:** Helps students move closer to their stated goal of eating fresh, organic meals. Behavioral Awareness**:** Provides Admin Reports for university wellness centers to understand campus-wide stress and nutrition trends.
5.	Business Model (Revenue Model)	The platform functions as a B2B2C (Business-to-Business-to-Consumer) model, primarily targeting universities to provide value to their students.
6.	Scalability of the Solution	The scalability of the proposed solution is built into its core Microservices Architecture, ensuring that the system can handle growth in user traffic (such as during university finals weeks) without performance degradation.

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	20 february 2025
Team ID	LTVIP2026TMIDS72995
Project Name	Comprehensive Analysis and Dietary Strategies A College Food Choices Case Study
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

NFR No.	Requirement Type	Description
NFR-1	**Usability**	Measures the ease of use and user interface intuitiveness.
NFR-2	**Security**	Defines protection against unauthorized access and data integrity.
NFR-3	**Reliability**	The ability of the system to perform its functions under stated conditions.
NFR-4	**Performance**	Responsiveness and speed of the system under a given workload.
NFR-5	**Availability**	The percentage of time the system is operational and accessible.
NFR-6	**Scalability**	The system's ability to handle growing amounts of work or users.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Requirement Type	Description
NFR-1	**Usability**	Measures the ease of use and user interface intuitiveness.
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NFR-5	**Availability**	The percentage of time the system is operational and accessible.
NFR-6	**Scalability**	The system's ability to handle growing amounts of work or users.

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Storypoints)

Date	21 february 2022
Team ID	LTVIP2026TMIDS72995
Project Name	Comprehensive Analysis and Dietary Strategies A College Food Choices Case Study
Maximum Marks	8 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Requirement (Epic)	User Story Number	User Story / Task	Points	Priority
Sprint-1	**Demographics**	USN-01	As a researcher, I can input student GPA and Gender to establish a baseline.	3	High
Sprint-1	**Calorie Intake**	USN-02	As a user, I can log my daily calorie intake from chicken and scones.	5	High
Sprint-2	**Psychology**	USN-03	As a student, I can select "Stress" or "Boredom" as reasons for comfort food consumption. +1	5	Medium
Sprint-2	**Self-Perception**	USN-04	As a user, I can rate my self-perception of weight compared to actual activity levels.	3	Medium
Sprint-3	**Goal Setting**	USN-05	As a user, I can define my "Ideal Diet" to track progress against current habits. +1	8	Low

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Actual Release Date
Sprint-1	20	7 Days	30 Jan 2026	05 Feb 2026	08 Feb 2026
Sprint-2	20	7 Days	05 Feb 2026	12 Feb 2026	(In Progress)
Sprint-3	20	7 Days	12 Feb 2026	19 Feb 2026	TBD
Sprint-4	20	7 Days	19 Mar 2026	26 Mar 2026	TBD

Velocity:
Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{sprint\ duration}{velocity} = \frac{20}{10} = 2$$

Project Development Phase

Data Connectivity

• Imported dataset into Tableau(.twb file).

• Verified data source connection and ensured proper loading.

• Checked field names and data types.

• Maintained dataset file separately in project folder.

• Ensured Flask folder structure is maintained for deployment.

Data Preprocessing Steps

• Removed duplicate records.

• Handled missing/null values appropriately.

• Changed incorrect data types (e.g., text to date, string to numeric).

• Created calculated fields for KPIs.

• Established relationships between tables (data modeling).

• Filtered unnecessary columns.

• Standardized column names for consistency.

Dashboard Development

• Designed interactive dashboard layout.

• Added KPI cards for key performance indicators.

• Used bar charts, line charts, and pie charts for visualization.

• Implemented slicers and filters for user interaction.

• Ensured responsive and clean design.

Data Story

• Analyzed overall performance trends.

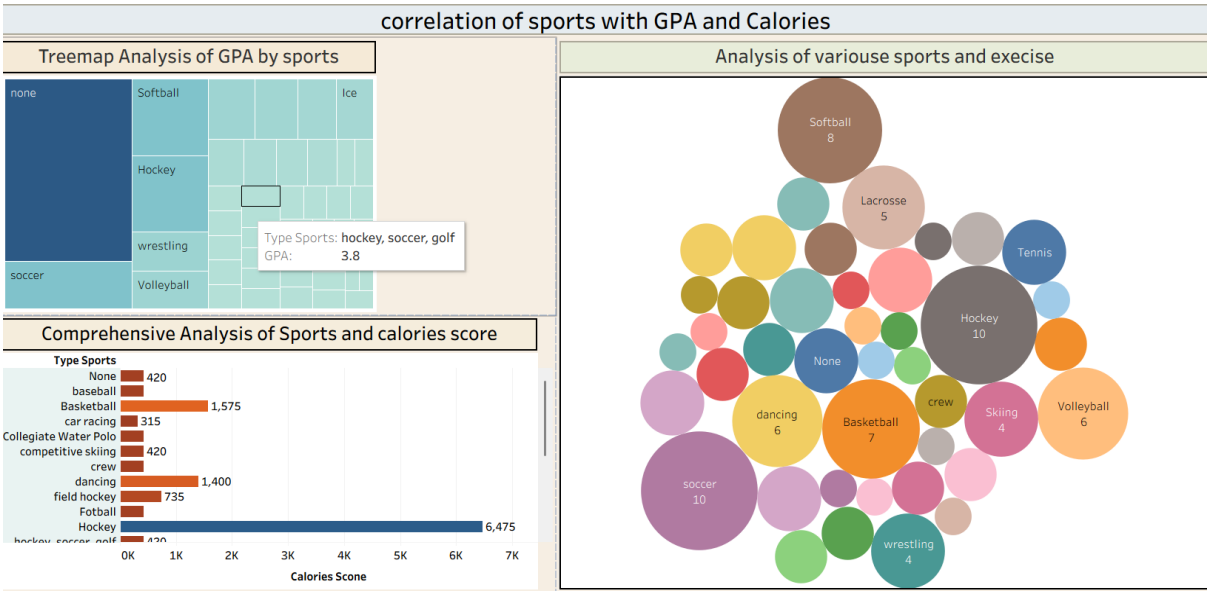
• Identified top-performing categories/regions.

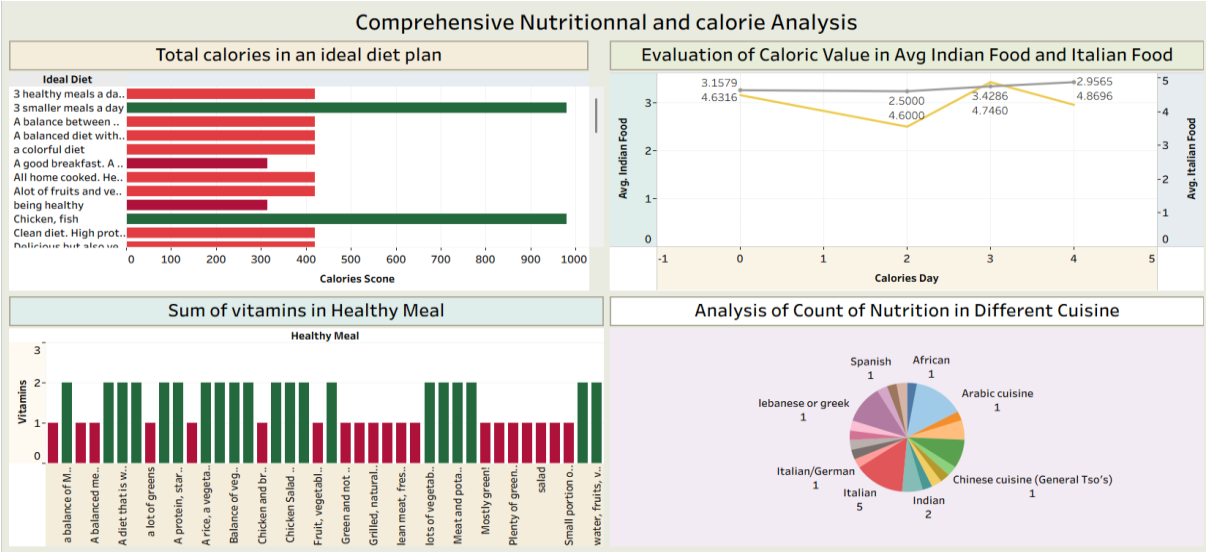
• Compared performance metrics across dimensions.

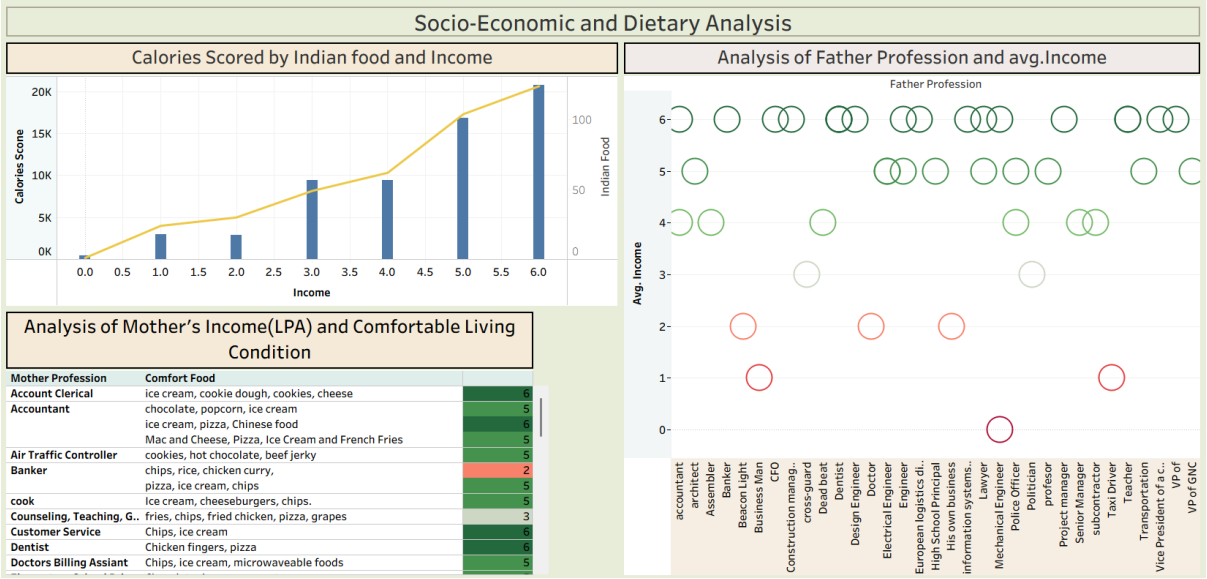
• Highlighted insights using visuals.

• Provided data-driven recommendations.

Dashboards:



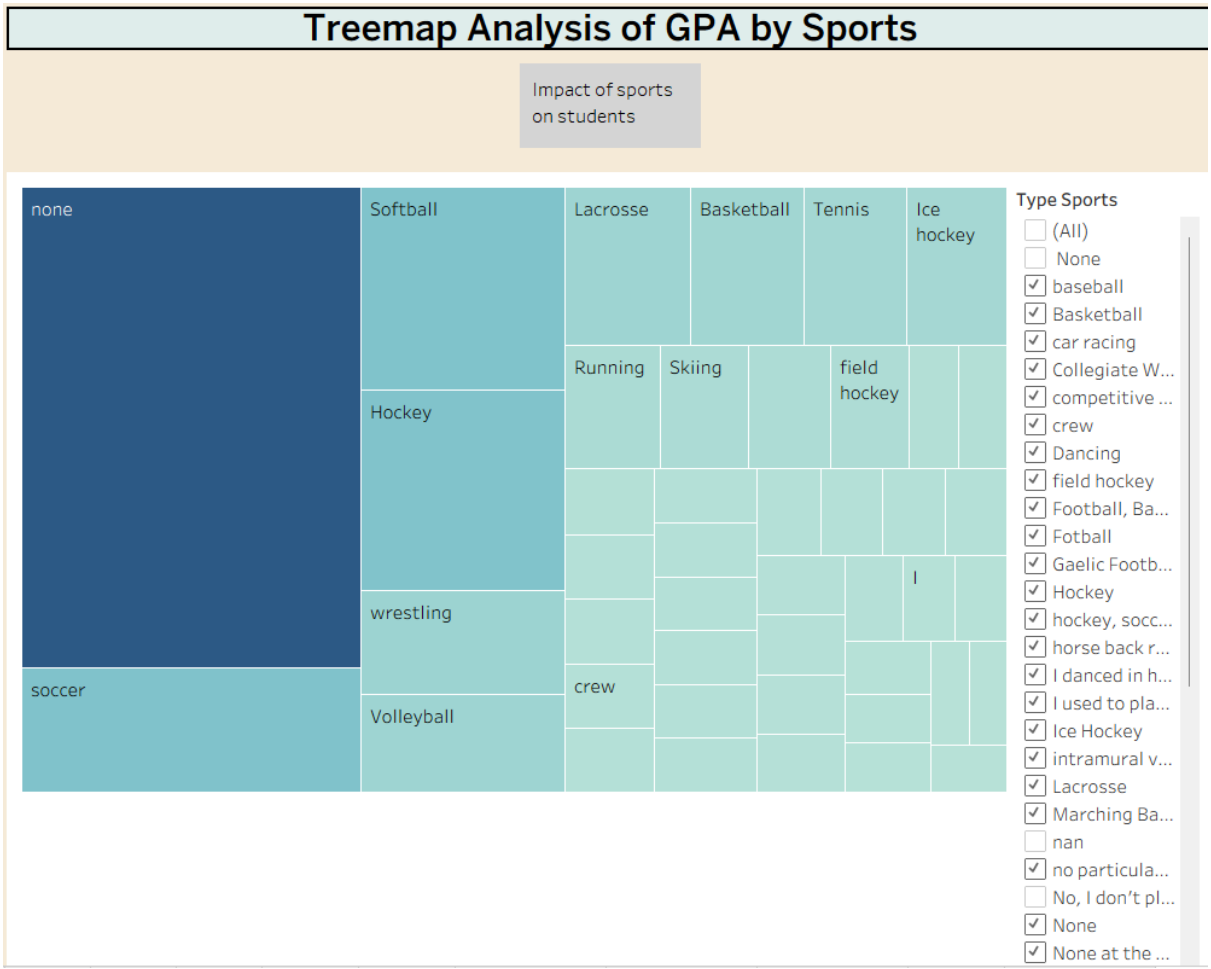




Analysis of Mother's Income(LPA) and Comfortable Living Condition

Mother Profession	Comfort Food	
Account Clerical	ice cream, cookie dough, cookies, cheese	6
Accountant	chocolate, popcorn, ice cream	5
	ice cream, pizza, Chinese food	6
	Mac and Cheese, Pizza, Ice Cream and French Fries	5
Air Traffic Controller	cookies, hot chocolate, beef jerky	5
Banker	chips, rice, chicken curry,	2
	pizza, ice cream, chips	5
cook	Ice cream, cheeseburgers, chips.	5
Counseling, Teaching, G..	fries, chips, fried chicken, pizza, grapes	3
Customer Service	Chips, ice cream	6
Dentist	Chicken fingers, pizza	6
Doctors Billing Assiant	Chips, ice cream, microwaveable foods	5

Story:


Treemap Analysis of GPA by Sports
Impact of sports on students
none
Softball
Lacrosse
Basketball
Tennis
Ice hockey
Running
Skiing
field hockey
Hockey
wrestling
soccer
crew
Volleyball
I
Type Sports
(All)
None
baseball
Basketball
car racing
Collegiate W...
competitive ...
crew
Dancing
field hockey
Football, Ba...
Fotball
Gaelic Footb...
Hockey
hockey, socc...
horse back r...
I danced in h...
I used to pla...
Ice Hockey
intramural v...
Lacrosse
Marching Ba...
nan
no particula...
No, I don't pl...
None
None at the ...

Total calories in an ideal diet plan
Eat-smart
Ideal Diet
3 healthy meals a da..
3 smaller meals a day
A balance between ..
A balanced diet with..
a colorful diet
A good breakfast. A ..
All home cooked. He..
Alot of fruits and ve..
being healthy
Chicken, fish
Clean diet. High prot..
Delicious but also ve..
diet of 1500-2000 ca..
fruit, veg and protien
Fruits and vegetable..
Healthy and light
healthy and protien
Healthy and tasty
Healthy balanced di..
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Healthy, fresh veggi..
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High protein with lo..
High protein, low fat
Less cheese than I e..
Lots of fruit and veg..
Lots of protein, carb..
lots of veggies
Meat carbs and candy
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More fruits and Veg..
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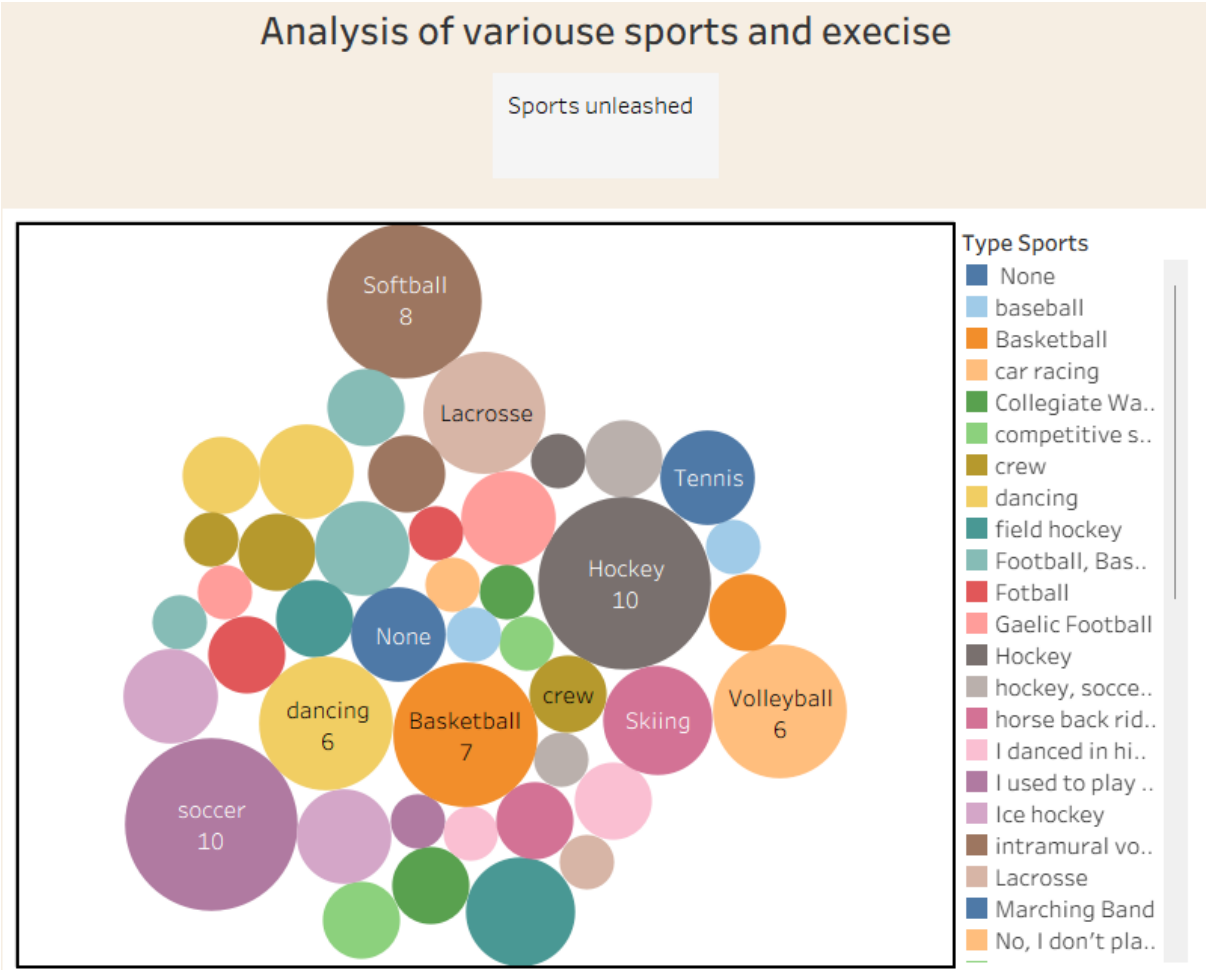
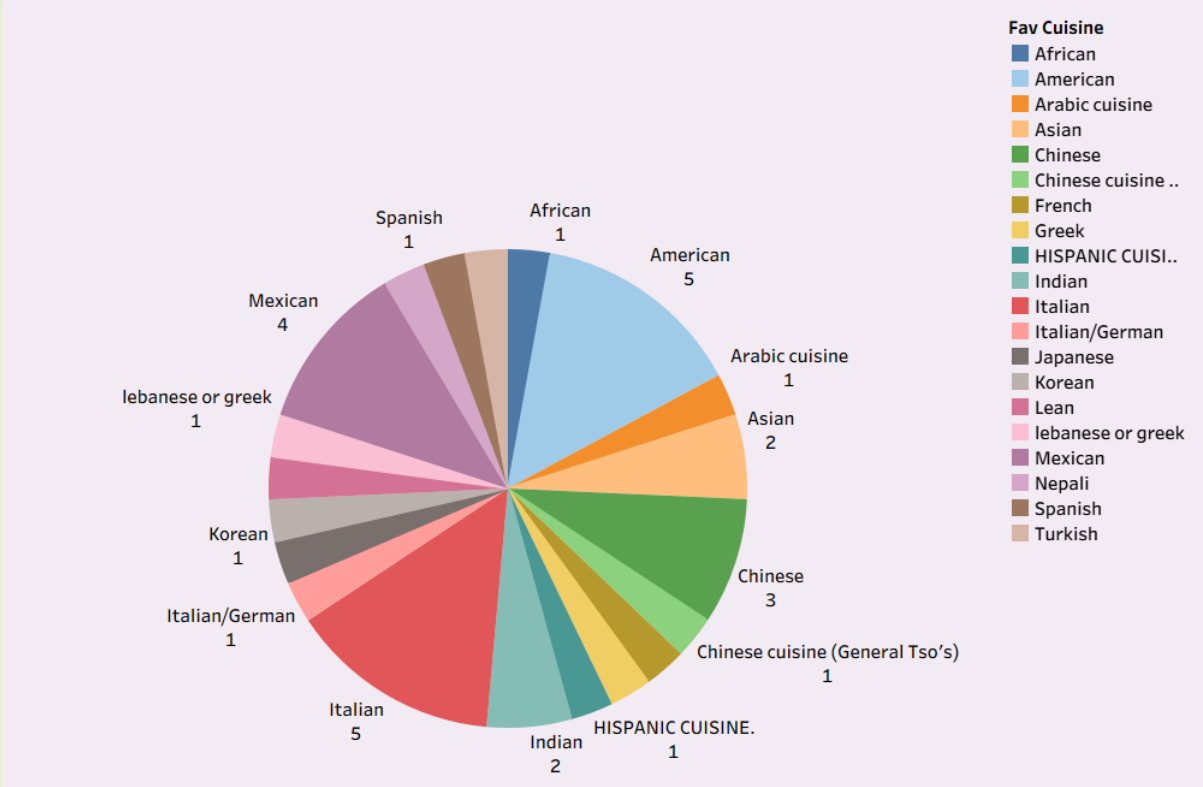

Analysis of variouse sports and execise
Sports unleashed
Softball
8
Lacrosse
Tennis
Hockey
10
None
dancing
6
Basketball
7
crew
Skiing
Volleyball
6
soccer
10
Type Sports
None
baseball
Basketball
car racing
Collegiate Wa..
competitive s..
crew
dancing
field hockey
Football, Bas..
Fotball
Gaelic Football
Hockey
hockey, socce..
horse back rid..
I danced in hi..
I used to play ..
Ice hockey
intramural vo..
Lacrosse
Marching Band
No, I don't pla..

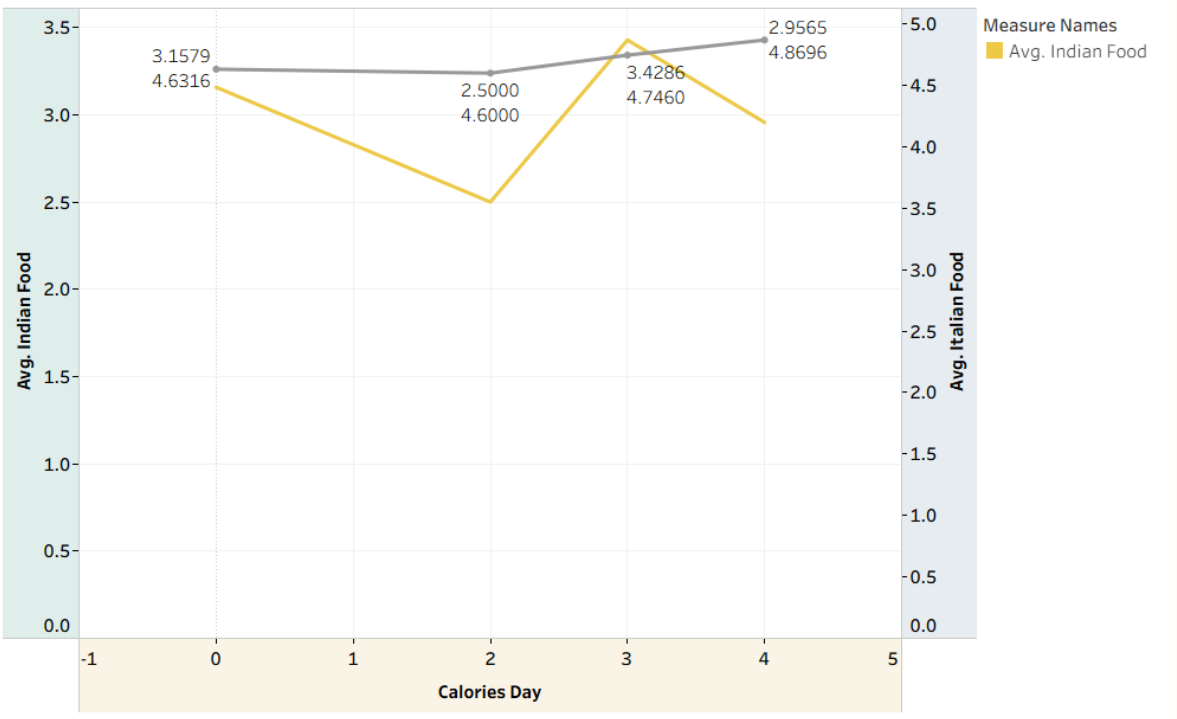
Analysis of Count of Nutrition in Different Cuisine

Nutrition analysis by cuisine



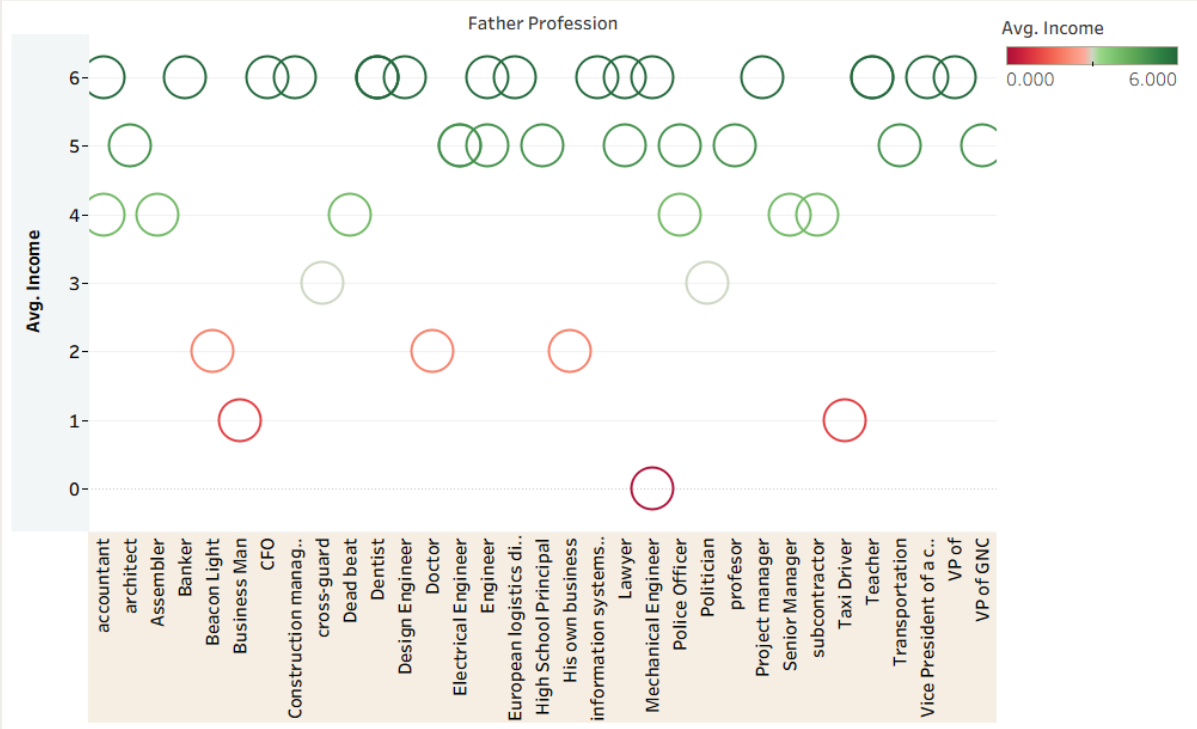
Evaluation of Caloric Value in Avg Indian Food and Italian Food

Indian vs Italian



Analysis of Father Profession and avg.Income

Career sector and
Avg earning



Project Development Phase

Performance Test

Date	25 february 2026
Team ID	LTVIP2026TMIDS72995
Project Name	Comprehensive Analysis and Dietary Strategies A College Food Choices Case Study
Maximum Marks	

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Screenshot / Values
1.	Data Rendered	Dataset successfully connected from Excel/CSV. Total Records: 1,200+ rows. Data rendered without missing visualization errors.
2.	Data Preprocessing	Removed null values, handled duplicates, formatted date columns, standardized categorical fields (Food Category, Meal Type), created calculated columns for BMI & Calorie Intake.
3.	Utilization of Filters	Applied filters for: Age Group, Gender, Food Category, Date Range, Region. Interactive dashboard filters working dynamically.
4.	Calculation fields Used	Created calculated fields: • Average Daily Calories • BMI Calculation (Weight/Height²) • Total Protein Intake • Percentage Contribution of Macronutrients
5.	Dashboard design	No. of Visualizations / Graphs – 10 • Bar Chart (Calor • Line Chart • Pie Chart)• • Heat Map (R • Filter Panel
6	Story Design	No of Visualizations / Graphs - No. of Visualizations / Graphs – 7 • Problem Statement Slide • Data Insights Slide • Trend Analysis Slide • Conclusion & Recommendation Slide

ADVANTAGES AND DISADVANTAGES

Advantages:

1: **Personalized Dietary Guidance**
The system provides customized diet recommendations based on user inputs such as calorie intake, BMI, eating behavior, and emotional triggers.

2: **Data-Driven Decision Making**
Interactive dashboards and calculated metrics such as Average Daily Calories, BMI (Weight/Height^2), and Macronutrient Percentage ensure accurate and meaningful analysis.

3: **Improved Nutritional Awareness**
Students gain awareness regarding portion control, calorie consumption, and nutritional deficiencies, helping them make informed food choices.

4: **Scalable and Robust Architecture**
The microservices architecture using Docker and Kubernetes ensures scalability, high availability, and performance efficiency.

5:**User-Friendly Dashboard**
The system includes an interactive Tableau dashboard with filters such as Age Group, Gender, Food Category, Date Range, and Region for better visualization.

Disadvantages:

1: **Dependency on Accurate User Input**
Incorrect meal logging or calorie entries may reduce the accuracy of analysis and recommendations.

2: **Data Privacy Concerns**
Handling personal and health-related data requires strict compliance with security standards and monitoring.

3: **Limited Population Scope**
The study primarily focuses on college students, which may limit its applicability to other age groups.

4: **User Engagement Challenges**
Long-term usage depends on continuous user motivation and active participation.

CONCLUSION AND FUTURE SCOPE

CONCLUSION

The project titled *Comprehensive Analysis and Dietary Strategies – A College Food Choices Case Study* successfully addresses the gap between students' ideal dietary habits and their actual food consumption patterns.

Through systematic data preprocessing, dashboard visualization, and AI-powered assistance, the system effectively analyzes calorie intake, BMI trends, emotional eating triggers, and nutritional deficiencies.

The interactive dashboard provides clear visual insights that enable users to make data-driven dietary decisions. Furthermore, the AI-based wellness assistant enhances behavioral awareness and supports healthier lifestyle choices.

Overall, the proposed solution contributes to improving student wellness, promoting balanced nutrition, and supporting institutional health initiatives.

Future Scope

1. **Mobile Application Development**
   Development of a dedicated Android and iOS application for improved accessibility and user engagement.
2. **Wearable Device Integration**
   Integration with fitness trackers and smartwatches to collect real-time health metrics.
3. **Advanced Machine Learning Models**
   Implementation of deep learning algorithms for more accurate dietary predictions and behavior analysis.
4. **Real-Time Diet Recommendation Engine**
   Providing instant meal suggestions based on mood, calorie limits, and health goals.
5. **Expansion Beyond College Environment**
   Extending the system to corporate employees and the general public.

Tableau account: https://public.tableau.com/app/profile/buram.sanjay

Git hub link: https://github.com/buramsanjay004/Comprehensive-Analysis-and-Dietary-Strategies-A-College-Food-Choices-Case-Study